

THE PENNY MAGAZINE



Issue 26

Monthly Supplement for August, 1832

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SUPPLEMENT OF THE PENNY MAGAZINE OF THE SOCIETY FOR
THE DIFFUSION OF USEFUL KNOWLEDGE, ISSUE 26, JULY 31 TO
AUGUST 31, 1832 ***

Monthly Supplement of
THE PENNY MAGAZINE

OF THE
Society for the Diffusion of Useful Knowledge.

26.]

JULY 31 TO AUGUST 31, 1832

STATUE OF JAMES WATT.



[Statue of James Watt, to be erected in Westminster Abbey.]

The people of this country have not been ungrateful to the memory of the great improver of the steam-engine. A bronze statue has been erected by subscription at Glasgow: another, of white marble, has been placed in the Hunterian Museum in the same large commercial town; and a third, also of white marble, which is completed, is about to adorn Westminster Abbey. From the latter statue the above drawing has been taken, by permission of Mr. Chantrey, who has been judiciously selected to execute these three important works. Independently of his high excellence as a sculptor, Mr. Chantrey had the advantage of modelling Watt some time before his death. Nothing can be finer in art, or more characteristic of the philosopher, than this statue. The perfect simplicity and repose of the composition, so expressive of the calm and reflecting mind of this extraordinary man, are beyond all praise.

The public meeting, at which it was determined to erect a statue to the memory of Watt, was perhaps one of the most interesting that ever was held in the metropolis. It took place on the 18th June, 1824; and the late Lord Liverpool, then Prime Minister, presided. A foreigner who was present, M. Dupin^[1], was particularly struck by the announcement from the first minister of the Crown, that the distinguished persons who met that day were assembled to pay a solemn tribute of gratitude and respect to the memory of an illustrious countryman. He says,—“The simple applications of industry, which, among the nations of antiquity, appeared unworthy of the study of philosophy, are now the final object of the most sublime theories. The mechanical arts, whose practice would have made the great geometers blush, who, like Archimedes, contributed to their advancement, have become a title of honour, in the nineteenth century, amongst the British nation.” It would be singular, indeed, if it were not so; and a minister of the Crown would be unfit for the government of our industrious community, if he did not feel that the great inventions which have grown out of our commercial superiority, and which have in a large degree created that superiority, were not eminently calculated to claim the noblest rewards that the people could bestow.

At this meeting Sir Humphry Davy moved the following resolution:—

“That the late James Watt, by his profound science, and by the original genius exhibited in his admirable inventions, has, more than any other of his contemporaries, demonstrated the practical utility of knowledge,

increased the power of man over the material world, and extended the comforts and enjoyments of human life.” This resolution was seconded by Mr. Boulton, the son of Mr. Watt’s distinguished partner.

The second resolution, that the services of James Watt to the civilized world demanded a national tribute of gratitude from his own country, was proposed by Mr. Huskisson, and seconded by Sir James Macintosh. We cannot forbear to quote a passage from the speech of this philosophical orator, whose words are now heard no more with reverence and delight: “In less than half a century, from the Mississippi to the Ganges, the name of Watt has been pronounced, and the benefits of his invention have been proved. If such a vast progress has been made in so small a number of years, what hopes may we not entertain of the future;—seeing that the combination of the useful and the fine arts has spread general information amongst such a multitude of minds—that knowledge has been placed within the reach of the humblest artisan—and that this class of men, for the most part remarkable for their intelligent, ingenious, active spirit, are full of the desire of instruction.”

The third resolution, that a monument to the memory of Watt should be erected in Westminster Abbey, was proposed by Mr. Brougham, and seconded by Mr. Peel. We cannot better close this notice, than with the concluding paragraph of Mr. Brougham’s speech:—“It is to honour the rare and excellent qualities of his character and genius, that we are assembled with the intention to erect a monument to the memory of the great engineer. Not that his memory has need of a monument to become immortal, for his name will last as long as the power which he has subjected to the use of man; but we are assembled to consecrate his example in the face of the universe—and to show to all our fellow-subjects that a man of extraordinary talent cannot better employ it than in rendering services to the human race. And where could we more fitly place the monument of this great man, than within a temple of that religion which preaches peace to all men, and instruction for the poor! The Pagan temples were decorated with the statues of warriors who had spread desolation amongst the people. Let ours be adorned with the statues of men who have contributed to the triumphs of Science and of Humanity; and above all by the statue of one who, without ever having given pain to any of his fellow-creatures, has been able to accomplish works which remain a lasting honour and benefit to society.”

1. Our brief notice of this meeting is taken from an account in the 'Discours sur l'Industrie,' &c., of this writer.

THE FEROE ISLANDS.

In the midst of a stormy sea, and under a sky rarely enlivened by sunshine, are situated the Feroe Islands. They are two-and-twenty in number, lying between the latitudes of $61^{\circ} 15'$ and $62^{\circ} 21'$, and extending altogether 61 miles in length from north to south, and 45 in breadth from east to west. Nothing can well be more rugged and desolate than their appearance. They consist of a crowded group of harsh steep rocks, the bases of which are separated by narrow and shallow channels of the sea, and whose sides, for the most part, are tremendous precipices, from two to three hundred fathoms high—complete walls of rock that strike the mariner with terror. Within these iron-bound coasts there are no valleys of any extent; but a few narrow dales, broken and craggy, are found between their summits, where the energies of man are hardly taxed to win a scanty supply from the soil which never exceeds four feet, and is frequently not more than eight inches in depth. This soil, moreover, is often swept away by the winds of the north, that in a moment can destroy the labour and improvement of years. These winds are indeed tremendous; they descend from the lofty hills to the shore, impelling large stones, and rolling them forward like balls; at times even tearing the turf from the sides of the hills, rolling it together like a sheet of lead, and precipitating it into the valleys. To hinder the roofs of their houses from being blown away, the poor Feroese are obliged to place boards upon them, and to throw ropes over them, the ends of which are fastened to heavy stones. Another dreadful enemy they have to contend with, is the voracious rat. These animals are so numerous, that they will sometimes destroy half a corn-field in a single night, and when they get at the sea-fowl in the crannies of the rocks, they commit such havoc that the bold fowler, who pursues his avocation at the risk of his neck, finds little left to reward

his pains and perils. In one island, the church-path, which is the best and often the only road, is so steep and narrow that at funerals the corpse is fastened to a board and carried upon men's shoulders. At another island, when the clergyman comes to visit them, which he does on Sundays as often as the weather and the sea will allow, they hoist him up the lofty steep cliffs, from his boat, by a rope, there being no other possible means of landing.

Yet in spite of these manifold difficulties and horrors, (according to Landt, a Danish pastor, who resided among them many years and who wrote an admirable account of these islands,) the Feroese are a virtuous, a civilized, and a happy people. Though obliged to struggle hard for their subsistence, which is mainly procured by the perilous occupations of fishing in their dark and stormy northern seas, and fowling among their giddy precipices, they all find time to acquire reading, writing, and arithmetic. There is not a school or a school-master in the whole group of islands; but parents instruct their own children. The worthy pastor says he found his parishioners remarkably fond of reading, well instructed in the Christian religion, and often thoroughly acquainted with the Bible. They make difficult calculations by head with facility, are good practical astronomers, and delight in the scientific game of chess, which many of them play admirably well. They are a sober people. In their long and dreary winter nights, instead of giving themselves up to the expensive and every-way pernicious excitement of drinking, they gather round their decent hearth with a book, which they read aloud by turns; or they play their favourite game, which by exercising their intellect also tends to their improvement. Whilst thus occupied, the storm may rage without—the furious ocean shake their rocky islets to the base—the night extend its long dark hours—the poor Feroese are happy in the comforts which their industry procures, and in the leisure they can so rationally employ.

INFANT SCHOOLS.

[Early Discipline Illustrated. By Samuel Wilderspin. Westley and Davis,
1832.]

This book possesses interest of no common kind. It contains an animated account of the progress hitherto made in the establishment of infant schools, and sets forth the advantages to be derived from their still farther extension.

The picture of infancy made happy by judicious means, is a most soothing subject for contemplation; made happy too, while being trained to virtuous habits and intellectual exertion, without having its buoyancy crushed by severity, or chilled by neglect. Mr. Wilderspin has here shown that an infant school, under good management, is not a scene of joyless restraint, but one of cheerful occupation. Here the infant learns to expand into an intellectual being, and to exercise all the kindly feelings of our nature. The author appears early to have directed his attention to the subject of education. He observes, "Every reflecting mind, perhaps, has some favourite topic of thought; and mine was directed, at an early period of life, to the instruction of the young. Naturally fond of children I pitied their state, both as altogether neglected and improperly treated under existing systems, and many schemes were revolved for the improvement of their condition. I observed them in the streets, in the dwellings of their parents, and in all the 'dame schools' to which I could gain access; thus the conviction increased, that a new order of things was most devoutly to be desired, and that it was urged on us alike by patriotism, philanthropy, and piety."

The systems of Bell and Lancaster, which were then coming into extensive operation, did not, in the opinion of Mr. Wilderspin, sufficiently embrace all the good that could be accomplished by education, since children are not placed in schools of that class until they are *seven* years old. Towards children under this age his care was therefore directed. In what measure he has succeeded in his benevolent task, a reference to his book will sufficiently evince. The mode in which Mr. Wilderspin has explained his views forcibly seizes upon the attention. The work before us is written in a playful, amiable spirit; and is replete with anecdotes, tending more to bring general conviction, than twenty learned lectures on the subject.

The good to be derived from educating children under seven years old is by some deemed chimerical, while by others the species of knowledge acquired is considered worse than useless. The most sceptical cannot, we think, continue in these opinions after a perusal of the work before us.

To acquire the ability of using the mental faculties is, in itself, a blessing which cannot be too highly prized; but this should be completely subordinate to the value of moral culture, and is but an accessory object of the infant school system. Upon this important point the author of this book thus distinctly expresses himself:—

“The *permanent* end and object is *moral* training, for which it is the first scholastic system which has made positive provision; indeed, which could effectually do so, seeing that it is the first attempt to mould the dispositions at that very tender age, when alone they can be moulded. But it is a transcendent discovery for the well-being of society, of which the present infants are to be the future men and women, that it has been demonstrated not only practicable, but easy to do so, from two to six years of age; and with 20,000 children it has already succeeded.”

The opinion of the present Bishop of London, as given in a recent charge, is thus quoted:—

“No person who has not himself watched them, can form an adequate notion of what these institutions, when judiciously conducted, may effect in forming the tempers and habits of young children; in giving them, not so much actual knowledge as that which, at their age, is more important, the habit and facility of acquiring it; and in correcting those moral defects which neglect or injudicious treatment would soon confirm, and render incurable. The early age at which children are taken out of our national schools, is an additional reason for commencing a regular and systematic discipline of their minds and wills, as soon as they are capable of profiting by it; and that is at the very earliest opening of the understanding.”

To originate and organize infant schools, on the best plan for imparting the greatest possible degree of good, appears to be the favourite occupation of Mr. Wilderspin. He is apparently possessed of a peculiar talent for attracting the attention of children, which eminently fits him for the task he has undertaken; while he possesses an energy in the cause, which no misfortunes can damp, or difficulties vanquish.

His wife, partaking of the same spirit, fell a sacrifice to the unremitting labour of superintending a school of two hundred children, without having any other assistance than her daughter, who was then a child. The narrative

of her illness and death, and of the affectionate grief of her young charge, is told with a simplicity and truth which cannot fail to reach the heart. All her numerous little pupils seemed to feel for her as if she were their parent; while this testimony of the success attendant on the culture of the kindly virtues, speaks more forcibly in favour of the system than any arguments which could be brought forward in its support. A short extract from this part of the volume will, perhaps, afford some idea of its touching interest.

“Aware that her end was approaching, she intimated this to her infant charge. Their replies discovered their warm affection: ‘We won’t part with you,’—‘We can’t let you go,’—‘We love you dearly,’—was often heard. When she told them of her exhaustion from so much talking and singing, some entreated that she would not work so hard; others promised to nurse her; and when she was confined to her room, many brought oranges, and all the little presents they could command. Indeed, in the whole course of my observation, I never met with ingratitude from children under six years of age; and how much they may be acted upon by love, those to whom reference is made amply proved.”

Subsequently Mr. Wilderspin travelled through the United Kingdoms, with the endeavour of establishing infant schools in every place where there was a sufficient population to warrant the hope of success. His glowing account of the advantages to be obtained from early education, and his persuasive arguments for its adoption, were often met by incredulity, or treated with ridicule. Undaunted by these obstacles, he determined to try the effect of practical proof, and to exhibit living witnesses of the actual benefit which might be obtained. Accordingly, in many places where arguments proved unavailing, he appealed to the more certain evidence of the senses. At one time we find him surrounded by groups of children, whom he was taking in carts from an organized school at some distance, that their proficiency might be exhibited to an unbelieving audience. The lively and graphic descriptions given of these scenes are calculated to amuse every class of readers, but they have much more than this to recommend them. It is delightful to view human nature under so pleasing an aspect. Here is no stern teacher governing school automata, drilled for exhibition, or terrified into industry, but a benevolent man, surrounded by happy beings, whom he is calling into intelligence and virtue, and whose minds and hearts he has awaked to every good impression.

In closing this book it is scarcely possible to resist the conviction, that infant education, pursued on an extended scale, is a most powerful

instrument for the amelioration of our species. The enthusiasm of the amiable writer must, we hope, in some measure be reflected on his readers, and his work may, in consequence, be of extensive utility, in commanding still more powerfully the attention of the public to the subject. Much still remains to be done.

NATURAL MAGIC.

[Letters on Natural Magic, addressed to Sir Walter Scott, Bart., by Sir David Brewster. Murray's Family Library. London, 1832. Price 5s.]

Natural Magic is the name given to those combinations of natural agencies, which, by the illusion or surprise which they produce, seem to us to possess supernatural power. Nor is the name an inappropriate one, though somewhat strange in sound. Magic, in the ordinary sense of the word, that is, supernatural power in human hands, exists only in the imagination; it is not a thing which has ever really been; it is a mere fancy, the offspring of ignorance and superstition, and nothing more. But if we can actually produce, by natural means, the same effects which the believers in magic say are to be achieved by the aid of spirits or other supernatural agents, we have a right to give the name of magic also to the art by which we do this—adding the epithet natural, to intimate that it is only the products of the magician's trickery which are imitated, and not his pretended mode of operation. If the impostor who professes to raise a spectre by a charm or incantation, calls the deception a piece of magic, the philosopher who does the same thing by an ingenious arrangement of mirrors, is still entitled to give it the same name.

On this subject Sir David Brewster has here produced a learned, instructive, and amusing work. The only regret or disappointment that the reader feels is on account of the parts of the subject which are only alluded to or slightly touched upon. The examples of the wonders of science given

in the present volume are only a selection from a much more abundant store of materials of the same kind. It is a selection, however, very judiciously made, and so as, if not to exhaust the subject, yet to present a view, more or less full, of each of its principal departments. First we have the illusions which affect us through the eye very largely treated of. The appearance of spectres to a brain or nervous system in a diseased or extraordinarily excited state,—the case of persons who are insensible to particular colours,—the tricks of the necromancers with concave mirrors,—the magic lantern and phantasmagoria,—the spectre of the Brocken, and the Fata Morgana,—are included, among many other things, under this head. The illusions depending on the ear, including the modern exhibition of the invisible girl, ventriloquism, the effects produced by the voice on glasses, the phenomena of echoes, &c. are considered in several of the following chapters. Then come two highly interesting chapters on mechanical feats and contrivances—such as remarkable exertions of strength, the automata of the ancients, Degennes's mechanical peacock, Vaucanson's duck, which ate and digested its food, Baron Kempelen's famous automaton chess-player, Duncan's tambouring machine, Babbage's calculating engine, &c. Lastly, the volume closes with a rapid survey of several of the most remarkable wonders of Chemistry. The art of breathing fire, that of walking upon red-hot iron, Aldini's incombustible dresses, the spontaneous combustion of human beings, the effects of the nitrous oxide, or laughing gas, and the very curious subject of certain elastic gases which the author himself has discovered in the cavities of gems, are some of the topics among which the reader is led on through this department.

What we have stated is enough to show what a fund both of amusement and of philosophy the book is. It is an excellent work for a mechanic's or village library.

The following is, we think, the most extraordinary of all the author's statements. The work, it is to be recollected, is addressed in the form of letters to Sir Walter Scott:—

One of the most remarkable and inexplicable experiments relative to the strength of the human frame, which you have yourself seen and admired, is that in which a heavy man is raised with the greatest facility, when he is lifted up the instant that his own lungs and those of the persons who raise him are inflated with air. This experiment was, I believe, first shown in England a few years ago by Major H., who saw it performed in a large party at Venice under the direction of an officer of the American Navy. As Major H. performed it more than once in my presence, I shall describe as nearly

as possible the method which he prescribed. The heaviest person in the party lies down upon two chairs, his legs being supported by the one and his back by the other. Four persons, one at each leg, and one at each shoulder, then try to raise him, and they find his dead weight to be very great, from the difficulty they experience in supporting him. When he is replaced in the chair, each of the four persons takes hold of the body as before, and the person to be lifted gives two signals by clapping his hands. At the first signal he himself and the four lifters begin to draw a long and full breath, and when the inhalation is completed, or the lungs filled, the second signal is given for raising the person from the chair. To his own surprise and that of his bearers, he rises with the greatest facility, as if he were no heavier than a feather. On several occasions I have observed that when one of the bearers performs his part ill, by making the inhalation out of time, the part of the body which he tries to raise is left as it were behind. As you have repeatedly seen this experiment, and have performed the part both of the load and of the bearer, you can testify how remarkable the effects appear to all parties, and how complete is the conviction, either that the load has been lightened, or the bearer strengthened by the prescribed process.

At Venice the experiment was performed in a much more imposing manner. The heaviest man in the party was raised and sustained upon the points of the fore fingers of six persons. Major H. declared that the experiment would not succeed if the person lifted were placed upon a board. He conceived it necessary that the bearers should communicate directly with the body to be raised. I have not had an opportunity of making any experiments relative to these curious facts; but whether the general effect is an illusion, or the result of known or of new principles, the subject merits a careful investigation.

Amongst the descriptions of mechanism calculated to excite popular curiosity, the following is very striking:—

One of the most popular pieces of mechanism which we have seen is the magician, constructed by M. Maillardet, for the purpose of answering certain given questions. A figure, dressed like a magician, appears seated at the bottom of a wall, holding a wand in one hand, and a book in the other. A number of questions ready prepared are inscribed on oval medallions, and the spectator takes any of these which he chooses, and to which he wishes an answer, and having placed it in a drawer ready to receive it, the drawer shuts with a spring till the answer is returned. The magician then rises from his seat, bows his head, describes circles with his wand, and consulting the book as if in deep thought, he lifts it towards his face. Having thus appeared to ponder over the proposed question, he raises his wand, and, striking with it the wall above his head, two folding doors fly open, and display an appropriate answer to the question. The doors again close, the magician resumes his original position, and the drawer opens to return the medallion. There are twenty of these medallions, all containing different questions, to which the magician returns the most suitable and striking answers. The medallions are thin plates of brass of an elliptical form, exactly resembling each other. Some of the medallions have a question inscribed on each side, both of which the magician answers in succession. If the drawer is shut without a medallion being put into it, the magician rises, consults his book, shakes his head, and resumes his seat. The folding-doors remain shut, and the drawer is returned empty. If two medallions are put into the drawer together, an answer is returned only to the lower one. When the machinery is wound up, the movements continue about an hour, during which time about fifty questions may be answered. The inventor stated, that the means by which the different medallions acted upon the machinery, so as to produce the proper answers to the questions which they contained, were extremely simple.

We give, in conclusion, the author's very just observations on the ultimate effect of inventions which at first sight appear to have no really useful object:—

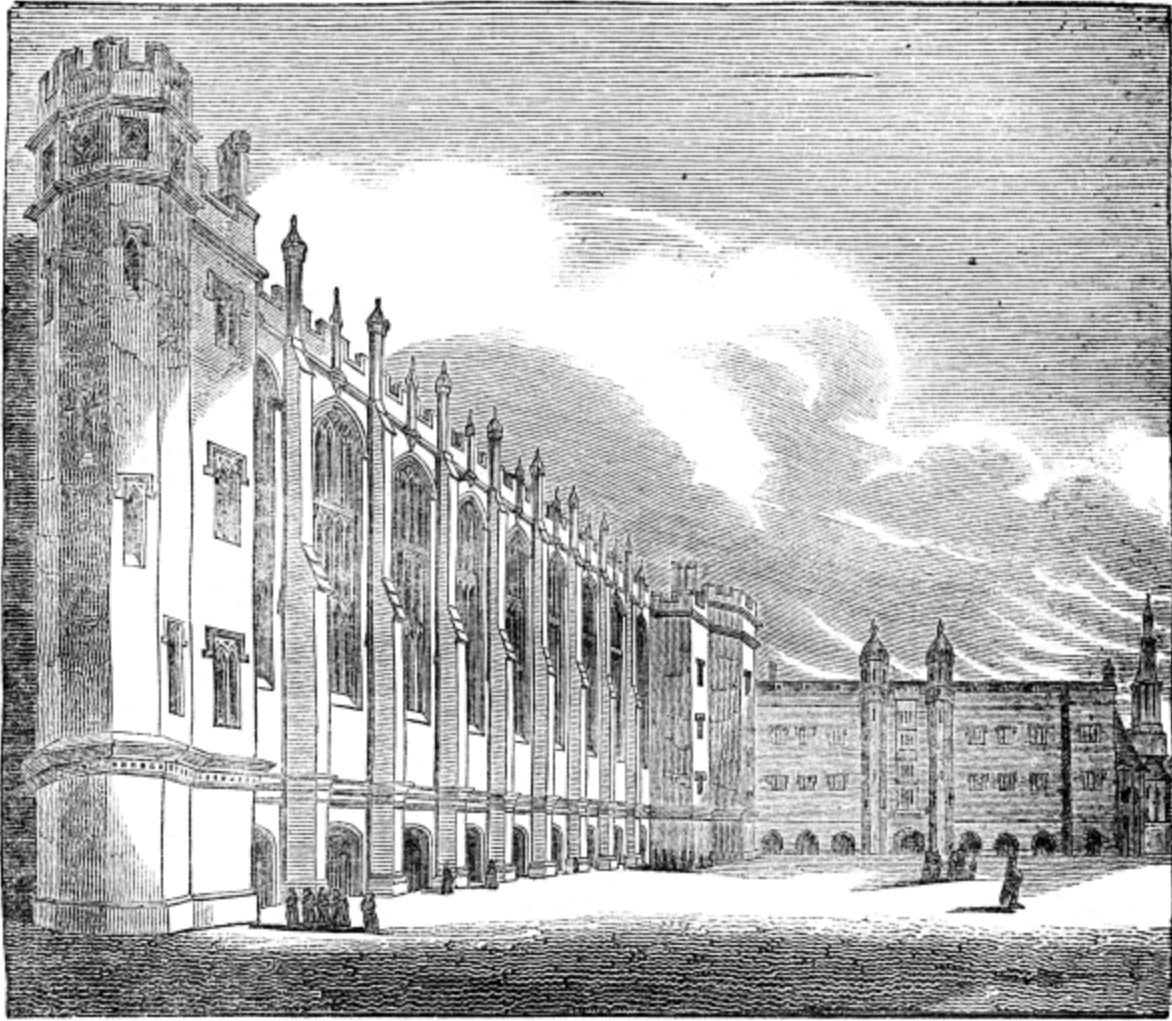
Ingenious and beautiful as all these pieces of mechanism are, and surprising as their effects appear even to scientific spectators, the principal object of their inventors was to astonish and amuse the public. We should form an erroneous judgment, however, if we supposed that this was the only result of the ingenuity which they displayed. The passion for automatic exhibitions which characterized the eighteenth century, gave rise to the most ingenious mechanical devices, and introduced among the higher orders of artists habits of nice and accurate execution in the formation of the most delicate pieces of machinery. The same combination of the mechanical powers which made the spider crawl, or which waved the tiny rod of the magician, contributed in future years to purposes of higher import. Those wheels and pinions, which almost eluded our senses by their minuteness, reappeared in the stupendous mechanism of our spinning-machines, and our steam-engines. The elements of the tumbling puppet were revived in the chronometer, which now conducts our navy through the ocean; and the shapeless wheel which directed the hand of the drawing automaton has served in the present age to guide the movements of the tambourine engine. Those mechanical wonders which in one century enriched only the conjuror who used them, contributed in another to augment the wealth of the nation; and those automatic toys which once amused the vulgar, are now employed in extending the power and promoting the civilization of our species. In whatever way, indeed the power of genius may invent or combine, and to whatever low or even ludicrous purposes that invention or combination may be originally applied, society receives a gift which it can never lose; and though the value of the seed may not be at once recognized, and though it may lie long unproductive in the ungenial soil of human knowledge, it will some time or other evolve its germ, and yield to mankind its natural and abundant harvest.

CHRIST CHURCH.

The recent additions to the buildings of the hospital consist of the Hall, and the Mathematical and Grammar Schools. The *Hall*, built under the superintendence of Mr. Shaw, whose sudden and melancholy death excited the public regret a few weeks ago, is constructed of Haytor granite, in the pure Gothic style of architecture, having an embattled summit, adorned with pinnacles, and flanked at each extremity of the front by an octagon tower. The southern side of this building is seen through an iron railing from Newgate-street. An open arcade, used as a play-ground for the boys in wet weather, forms the lower story. The dining-hall occupies the upper

story, and is a noble apartment, calculated to accommodate eight hundred scholars. It is 157 feet in length by 52 in breadth, and 47 in height. At the eastern end is a large organ, used to lead the hymns of the children. The western end is occupied by a gallery for the use of visitors; and against the north wall a pulpit is erected for prayer and exhortation. The walls are decorated by several fine pictures by Holbein, Verrio, and others, illustrative of the history of the place. To this hall the public are admitted by tickets every Sunday evening from the first Sunday in February to Easter, to witness the *public suppers*, as they are called, of the boys. Tickets may be obtained from any of the governors; and the sight is an extremely interesting one. The building was opened for public use on the 19th of May, 1829.

The *Mathematical* and *Grammar Schools* are contained in one building; of which the cut represents the southern front. It is constructed of yellowish brick with stone facings, in what is called the Tudor style of architecture, and from the designs of Mr. Shaw. A statue of Charles the Second ornaments the eastern wing. This building was opened for public use on Easter, 1832. The expense of the new hall, schools, and wards, are understood to be about £30,000.



[The Dining-Hall.]



[The Mathematical and Grammar Schools.]

BRITISH INDIA.

[Historical and Descriptive Account of British India (in the Edinburgh Cabinet Library). 3 vols. 12mo. Edinburgh, Oliver and Boyd, 1832. Price

To the imagination of the people of Europe, for more than two thousand years, a peculiar charm has hung around the name of India. Its remoteness and separation from our quarter of the globe; its high, and at the same time anomalous civilization; its singular and fantastic institutions of civil polity; its mysterious, stern, and imposing religion; its dazzling abundance of gold and precious gems, and other productions, which have usually most inflamed the cupidity of men; its sunny sky and teeming soil, and its swarming population,—all these considerations long threw over India the colours of a land seen in a dream, or read of in some volume of romance or poetry. Nor when, at last, commerce had made a high road to it over the waves, and for the narratives, filled with unexplained wonders, of individual voyagers, we had fleets constantly arriving, loaded with its merchandize, did it cease still to dwell in the public mind as a splendid vision. In many respects the nearer and more distinct view only disclosed new features of magnificence and attraction. From a foreign country, with which we carried on trade by means of some two or three marts, established here and there on the border of the immense peninsula, India has become, throughout nearly its whole extent, a part of our own empire—a subjugated and tributary possession of our little island. This is the most extraordinary of all the revolutions which have marked the history of that portion of the earth, and, together with the state of things which has grown out of it, the political and social systems which have sprung up under the British connexion, and the changes through which both are likely to pass, gives an interest to the subject of India greatly exceeding any that before surrounded it.

The present publication, as its general title, which we have quoted, indicates, is an account of British India, considered not only historically, but in its actual condition. It is the work of several writers, some of whom have long been eminent for their acquaintance with the subjects which they here undertake to illustrate. Only the first volume, and somewhat more than the half of the second, is devoted to what may be properly called the history of India; but the remainder of the second volume, which appears to be contributed by the same writer, Mr. Hugh Murray, consists of five chapters—on the Hindoo History and Mythology; on the Hindoo Manners and

Literature; on the British Government of India; on the British Social System in India; and on the Industry and Commerce of India: which may be considered as belonging to the same portion of the work. These two volumes are written in a flowing, perspicuous, and agreeable style, and will be found to present a very interesting and useful outline of the succession of great events, of which India has been the theatre since it was overrun by the first swarm of its western invaders. The principal fault we have to find with the narrative is, that the notices of important transactions are occasionally so brief as to be unsatisfactory, if not almost unintelligible, and that this unpleasant effect seems, in some degree, to be occasioned by an over anxiety on the part of the author to preserve the mere smoothness of his style, by the omission sometimes even of a necessary clause, or sentence of explanation, which might have interrupted the ease or equability of its current. In other instances, the brevity with which parts of the story are treated may be imputable solely to the limited space which the author had at his command; but the effect is sometimes awkward enough. The only account, for example, which we have of the late contest with the Birmese, is contained in the following two sentences, the second of which, by the bye, would seem almost to contradict the first: “Since the termination of the Pindarree contest, no important event or acquisition has distinguished the history of British power in Indostan. The only war undertaken on a great scale was the arduous but finally successful one with the Birman empire, by which the Company gained a considerable territory along the Bay of Bengal.”—(ii. 283.) We would also recommend to Mr. Murray a more liberal insertion of dates than we usually find in his historical compilations. The year in which an event happened can always be given without much expenditure of space; and nothing contributes more to the clearness of such narratives as the present.

The third volume is the work of various authors. The Zoology of India is treated of in the first five chapters, by Mr. James Wilson, who is well known for his familiarity with the subject of Natural History, and whose descriptions are in general lively and graphic, while he appears to have exerted a very laudable industry in the collection of his materials from the best sources. Two chapters on Indian Botany follow from the pen of Dr. Greville, who has had the advantage of some valuable communications from Dr. Wallich, the able and zealous superintendent of the Botanic Garden at Calcutta. The climate, the hydrography, the geology, and

mineralogy of India, are then described by the learned pen of Professor Jameson; after which comes a chapter on the general medical effects of the Indian climate by Dr. Ainslie, late of the Medical Staff of Southern India; and another on the Spasmodic Cholera, as it appeared in that country, by Mr. Rhind. A popular view of the Hindoo astronomy and mathematics, followed by an account of some interesting trigonometrical surveys executed by the late Colonel Lambton, is then given by Professor Wallace; and the volume concludes with two interesting chapters by Captain Clarence Dalrymple, in which that officer describes the manner in which the navigation to India is at present performed, and discusses the question as to the best means of establishing a steam communication between that country and England.

We regard this publication as extremely well adapted to introduce its readers to an acquaintance with India. In respect to many matters it is fitted indeed rather to stimulate to further inquiry than to afford complete satisfaction; but this, which is an unavoidable consequence of the limited size of the work, we do not look upon as a defect in an elementary book. Whatever the subject, the first view of it which is presented to the mind of the student ought to be a general one, for the plain reason that it can in no otherwise be a comprehensive one.

THE HYDROSTATIC BED FOR INVALIDS.

We are favoured by Dr. Arnott, the author of ‘The Elements of Physics,’ with an unpublished extract from the fifth edition of his work, now in the press. The invention here described promises to be such a real blessing to humanity, that we feel great pleasure in assisting to make known an improvement of such importance in the healing art; particularly as its value is not a matter of speculation, and as its benevolent author freely allows its use wherever the wants of his fellow-creatures require its application:—

In many of the diseases which afflict humanity, more than half of the suffering and danger is not really a part of the disease, but the effect or consequence of the confinement to which the patient is subjected. Thus a fracture of the bone of the arm is as serious a local injury as a fracture of one of the bones of the leg; but the former leaves the patient free to go about and amuse himself, or attend to business as he wills, and to eat and drink as usual—in fact, hardly renders him an invalid; while the latter imprisons the patient closely upon his bed, and brings upon him, first, the irksomeness of the unvaried position, and then the pains of the unequal pressures borne by the parts on which the body rests. These, in many cases of confinement, disturb the sleep and the appetite, and excite fever, or such constitutional irritation as much to retard the cure of the original disease, and not unfrequently to produce new and more serious disease. That complete inaction should prove hurtful to the animal system, may by all be at once conceived; the operation of the continued local pressures will be understood from the following statements. The health, and even life, of every part of the animal body depends on the sufficient circulation through it of fresh blood, driven in by the force of the heart. Now when a man is sitting or lying, the parts of his flesh compressed by the weight of the body do not receive the blood so readily as at other times; and if from any cause the action of his heart has become weak, the interruption will both follow more quickly and be more complete. A peculiar uneasiness soon arises where the circulation is thus obstructed, impelling the person to change of position; and a healthy person changes as regularly, and with as little reflection, as he winks to wipe and moisten his eye-balls. A person weakened by disease, however, while he generally feels the uneasiness sooner, as explained above, and therefore becomes what is called restless, makes the changes with much fatigue; and should the sensations after a time become indistinct, as in the delirium of fever, in palsy, &c., or should the patient have become too weak to obey the sensation, the compressed parts are kept so long without their natural supply of blood that they lose their vitality, and become what are called sloughs or mortified parts. These have afterwards to be thrown off, if the patient survive, by the process of ulceration, and they leave deep holes, requiring to be filled up by new flesh during a tedious convalescence. Many a fever, after a favourable crisis, has terminated fatally from this occurrence of sloughing on the back or sacrum; and the same termination is common in lingering consumptions, palsies, spine diseases, &c., and generally in diseases which confine the patients long to bed.

It was to mitigate all, and entirely to prevent some, of the evils attendant on the necessity of remaining in a reclining posture, that the hydrostatic bed was contrived. It was first used under the following circumstances.

A lady after her confinement, which occurred prematurely, and when her child had been for some time dead, passed through a combination and succession of low fever, jaundice, and slight phlegmasia dolens of one leg. In her state of extreme depression of strength and of sensibility, she rested too long in one posture, and the parts of the body on which she had rested all suffered: a slough formed on the sacrum, another on the heel; and in the left hip, on which she had lain much, inflammation began, which terminated in abscess. These evils occurred while she was using preparations of bark, and other means, to invigorate the circulation, and while her ease and comfort were watched over by the affectionate assiduity of her mother, with numerous attendants. After the occurrence, she was placed upon the bed contrived for invalids by Mr. Earle, furnished for this case with pillows of down and of air of various sizes, and out of its mattress portions were cut opposite to the sloughing parts; and Mr. Earle himself soon afforded his valuable aid. Such, however, was the reduction of the powers of life, that in spite of all endeavours, the mischief advanced, and about a week later, during one night, the chief slough on the back was much enlarged, another had formed near it, and a new abscess was produced in the right hip. An air-pillow had pressed where the sloughs appeared. The patient was at that time so weak that she generally fainted when her wounds were dressed; she was passing days and nights of uninterrupted suffering, and as all known means seemed insufficient to relieve her, her life was in imminent danger.

Under these circumstances, the idea of the hydrostatic bed occurred to me. Even the pressure of an air-pillow had killed her flesh, and it was evident that persons in such a condition could not be saved unless they could be supported without sensible inequality of pressure. I then reflected, that the support of water to a floating body is so uniformly diffused, that every thousandth of an inch of the inferior surface has as it were its own separate liquid pillar, and no one part bears the load of its neighbour—that a person resting in a bath is nearly thus supported—that this patient might be laid upon the surface of a bath over which a large sheet of the water-proof India-rubber cloth were previously thrown, she being rendered sufficiently buoyant by a soft mattress placed beneath her—thus would she repose on the face of the water, like a swan on its plumage, without sensible pressure anywhere, and almost as if the weight of her body were annihilated. The pressure of the atmosphere on our bodies is of fifteen pounds per square inch of its surface, but, because uniformly diffused, is not felt. The pressure of a water-bath of depth to cover the body, is less than half a pound per inch, and is similarly unperceived. A bed such as then planned was immediately made. A trough of convenient length and breadth and a foot deep was lined with metal to make it water-tight; it was about half filled with water, and over it was thrown a sheet of the India-rubber cloth as large as would be a complete lining to it if empty. Of this sheet the edges, touched with varnish to prevent the water creeping round by capillary attraction, were afterwards secured in a water-tight manner all round to the upper border or top of the trough, shutting in the water as closely as if it had been in bottles, the only entrance left being through an opening at one corner, which could be perfectly closed. Upon this expanded dry sheet a suitable mattress was laid, and constituted a bed ready to receive its pillow and bed-clothes, and not distinguishable from a common bed but by its most surpassing softness or yielding. The bed was carried to the patient's house, and she was laid upon it; she was instantly relieved in a remarkable degree: sweet sleep came to her; she awoke refreshed; she passed the next night much better than usual; and on the following day, Mr. Earle found that all the sores had assumed a healthy appearance: the healing from that time went on rapidly, and no new sloughs were formed. When the patient was first laid upon the bed, her mother asked her where the down pillows, which she before had used, were to be placed; to which she answered, that she knew not, for that she felt no pain to direct: in fact, she needed them no more.

It may be here recalled to mind, that the human body is nearly of the specific gravity of water, or of the weight of its bulk of water, and therefore, as is known to swimmers, is just suspended or upheld in water without exertion, when the swimmer rests tranquilly on his back with his face upwards. He then displaces water equal to his own body in weight as well as in bulk, and is supported as the displaced water would have been. If his body be two and a half cubical feet in bulk (a common size), he will just displace two and a half cubic feet of water, equal in weight to his body. If, however, instead of displacing the water with his mere body, he choose to have something around or under him which is bulky with little weight, as the mattress of the bed above described,—when his weight has forced two cubical feet of that under the level of the water around, he will float with four-fifths of his body above the level, and will sink much less into his floating mattress than a person sinks in an ordinary feather-bed. It thus appears that by choosing the thickness of the mattress, and if unusual positions are required, by having different thickness in different parts, or by placing a bulk of folded blanket or of pillow over or under the mattress in certain situations, any desired position of the body may be easily obtained.

This bed is a warm bed, owing to water being nearly an absolute non-conductor of heat from above downwards, and owing to its allowing no passage of cold air from below. From this last fact, however, less of the perspiration, whether sensible or insensible, will be carried off by the air than in a common bed, and unless the patient can leave the bed daily to let it be aired, it is necessary to lay an oiled silk or other water-proof cloth over the mattress, to prevent the perspiration from descending to be condensed on the cloth below; or to place a blanket below to be changed occasionally; or, finally, to lay under the mattress a layer of cork, cut into small pieces, and so connected as to leave passages between, for any desired degree of ventilation. This bed is in itself as dry as a bed can be,

for the India-rubber cloth (of which bottles can be made) is quite impermeable to water, and the maker is now preparing cloth expressly for this purpose. Then, as Sir Humphry Davy recommended that his safety lamp should be double, some persons may prefer a double sheet, to obviate the possibility of accident. Unlike any other bed that ever was contrived, it allows the patient, when capable of only feeble efforts, to change his position, almost like a person swimming, and so to take a degree of exercise, affording the kind of relief which in constrained positions is obtained by occasional stretching, or which an invalid seeks by driving out in a soft-sprunged carriage. It exceedingly facilitates turning for the purpose of dressing wounds, for by raising one side of the mattress or depressing the other, or merely by the patient's extending a limb to one side, he is gently rolled over, nearly as if he were simply suspended in water; and it is possible even to dress wounds, apply poultices, or place vessels under any part of the body, without moving the body at all; for there are some inches of yielding water under the body, and the elastic mattress may at any part be pushed down, leaving vacant space there, without the support being lessened for the other parts. Then, with all the advantages which other invalid beds possess, and with those which are entirely its own, it may yet be made so cheaply, that even in hospitals, where economy must prevail, it may at once be adopted for many of the bed-ridden. Mr. Earle, within a few days of seeing the first one, had others made for patients in St. Bartholomew's Hospital, and has been as much pleased with the results of them as of the first. The bed has since been introduced into St. George's Hospital by Mr. Keate, and elsewhere.—The author has now seen enough of the effects of this bed to make him feel it a duty at once to publish a notice of it. With it, evidently, the fatal termination called sloughing, now so common, of fevers, and other diseases, need never occur again. And not only will it prevent that termination, but by alleviating the distress through the earlier stages, it may prevent many cases from even reaching the degree of danger. Then it is peculiarly applicable to cases of fractured bones, and other surgical injuries; to palsies, diseases of the hip joint, and spine; and universally where persons are obliged to pass much time in bed. And in all classes of curvature of the spine, either actually existing or threatened, it affords a means of laying a patient in any desired position, and with any degree of pressure incessantly urging any part of the spine back to its place. If used without the mattress, it becomes a warm or a cold bath, not allowing the body however to be touched by the water; and in India, it might be made a cool bed for persons sick or sound, during the heats which there prevent sleep and endanger health. There are numerous other professional adaptations and modifications of it, which will readily occur to practitioners sufficiently versed in the department of natural philosophy (hydrostatics) to which it belongs. Before reflection a person might suppose a resemblance between it and an air-bed or pillow, calling this a water-bed or pillow; but the principles of the two are perfectly distinct or opposite. An air-pillow supports by the *tension of the surface* which encloses the air, and is therefore like a hammock or the tight sacking under the straw mattress of a common bed, and really is a hard pillow; but in the hydrostatic bed there is no tense surface or web at all: the patient is floating upon the water, on which a loose sheet is lying, merely to keep the mattress dry, and every point of his body is supported by the water immediately beneath it. To recall the difference here described, and which is of great importance, the bed is better described by the appellation of *hydrostatic bed* than of *water-bed*.

The author has given no exclusive right or privilege to any person to make this bed. He has hitherto employed the carpenter nearest to him, Mr. Smith, 253, Tottenham-court Road, at the back of Bedford Square, and the manufacturers of the water-proof cloth, Mackintosh and Co., 58, Charing-Cross; but any carpenter or upholsterer may learn to supply them, and he gives free permission to all.

The preceding paragraphs are intended as much to direct in the choice and use of common beds for the sick, as to announce and describe the hydrostatic bed for the cases where it may be required. At present the

medical attendant often leaves whatever regards the bed to the judgment of friends or nurses; but evidently he who perceives clearly how much the course and events of a malady may depend on the patient's being supported, so that no pain shall arise from local pressure, and as little weariness as possible from the constrained position, will be better able to use any bed to its greatest advantage, and where a choice of beds is allowed, to choose the best. There is a bed formed of spiral springs which diffuses the support more than any bed except the hydrostatic bed; and had professional men generally known it, it would have been more extensively used than it is, and would have received modifications of which it is susceptible for medical purposes. It has been long known, but chiefly as a mechanical curiosity or as an object of luxury,—but so little known, that a few years ago an English manufacturer took out a patent for it, believing it a novelty. It is now made by upholsterers generally, and the same principle is applied in the construction of sofas, chairs, and carriage cushions.

It is one of the many striking facts illustrative of the benefits attending the commercial intercourse of mankind, that India-rubber, or caoutchouc, for many years used only to rub out pencil-marks, has become an important article in the manufacture of cloth; and it is pleasing to think that probably one of its most extensive applications will be in the mitigation of human suffering.



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- [p. 212](#): Replaced comma with period after phrase “the experiment would not succeed if the person lifted were placed upon a board.”
- [p. 212](#): Replaced comma with period after phrase “thin plates of brass of an elliptical form, exactly resembling each other.”
- [p. 212](#): Replaced “fo ding-doors” with “folding-doors” in phrase “The folding-doors remain shut, and the drawer is returned empty.”
- [p. 212](#): Added period after phrase “seen through an iron railing from Newgate-street.”
- [p. 215](#): Added period after phrase “because uniformly diffused, is not felt.”

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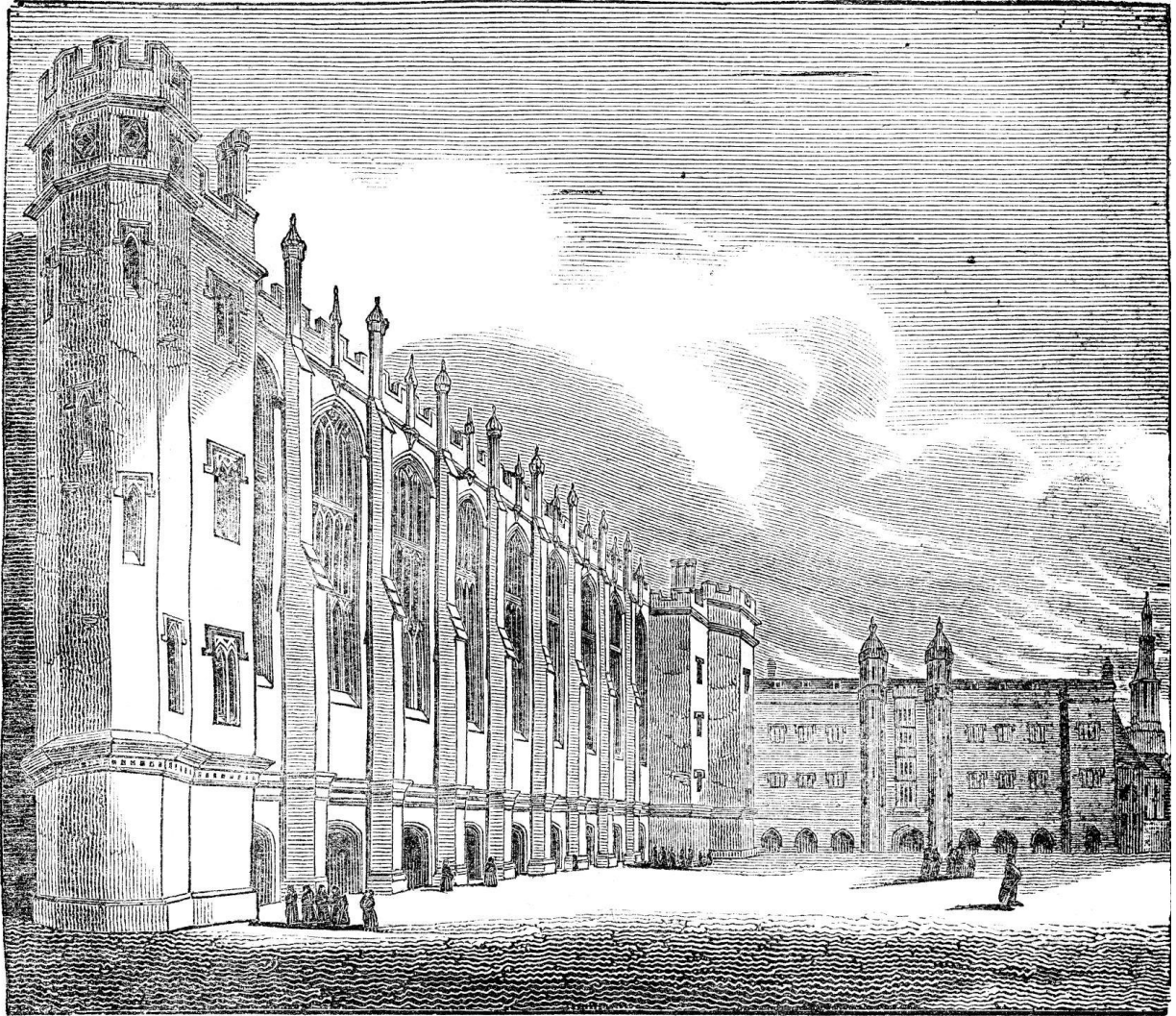
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